

Inceptez Data Science & AI – Industry Project

Objective

The purpose of this project is to simulate the type of assignment a Data Scientist may receive after joining an organisation.

You are not solving a textbook ML problem.

You are part of the company's Data Science team. A business department has approached your team with a problem. Your responsibility is to:

- understand the business requirement;
- translate the requirement into a Machine Learning problem;
- understand and prepare the data;
- identify the target variable and predictors;
- build appropriate classification models;
- evaluate the models from both technical and business perspectives;
- recommend a model;
- explain how the company could use the model.

Models permitted

For this project, use only techniques covered in class so far:

- Logistic Regression
- K-Nearest Neighbours (KNN)
- Decision Tree

You are expected to experiment with all three models and recommend one final approach.

PROJECT 1 — Customer Campaign Response Prediction

Company

NorthStar Bank – Customer Analytics Division

NorthStar Bank offers savings accounts, loans, credit products and investment products to its customers.

One of the bank's products is a **Term Deposit**.

At present, the bank's marketing team contacts a large number of customers by telephone whenever a Term Deposit campaign is launched.

Calling every customer has several disadvantages:

- high call-centre cost;
- customers receiving irrelevant calls;
- significant employee time spent on customers unlikely to purchase;
- lower productivity of the marketing team.

The Marketing Head has approached the Data Science team for assistance.

Business Requirement

The bank would like to use historical campaign information to identify customers who are more likely to subscribe to a Term Deposit.

Before starting the next campaign, the marketing team wants the Data Science team to provide a model that can classify customers into:

Likely to Subscribe

or

Unlikely to Subscribe

The marketing team can then prioritise customers predicted to have a greater likelihood of subscribing.

Your Role

You are a Data Scientist in the **Customer Analytics Team**.

Your manager has asked you to investigate whether historical customer and campaign information can be used to predict customer response.

Dataset

Use the:

UCI Machine Learning Repository – Bank Marketing Dataset

Dataset ID: **222**

The dataset contains information collected during direct-marketing campaigns of a Portuguese banking institution.

The target variable is:

y

where:

- yes = customer subscribed to the Term Deposit
- no = customer did not subscribe

The complete dataset contains approximately 45,000 customer records and includes both numerical and categorical variables.

Important Business Constraint

The dataset contains a variable called:

`duration`

This represents the duration of the telephone call.

Think carefully before using this variable.

The business wants to predict which customer should be contacted **before the call takes place**.

At that point, the duration of the future call cannot possibly be known.

Therefore, your production model should **not use duration**.

This is an example of **data leakage**.

Part of your responsibility as a Data Scientist is not simply to use every column available in a dataset, but to determine whether that information would actually be available when the prediction is made.

Project Requirements

1. Understand the Business Problem

Explain in your own words:

- What problem is the marketing team trying to solve?
- Why would an ML model be useful?

- What exactly are we predicting?
 - Is this regression or classification?
 - Is this binary or multiclass classification?
 - What would the bank do with the prediction?
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2. Understand the Data

Perform initial data exploration.

At minimum, investigate:

- number of rows and columns;
- feature names;
- data types;
- numerical variables;
- categorical variables;
- missing/unknown values;
- duplicate records;
- target variable distribution.

Identify the percentage of customers who subscribed and who did not subscribe.

Comment on whether the classes appear balanced.

3. Exploratory Data Analysis

Investigate relationships that may be useful to the business.

Examples include:

- subscription rate by age group;
- subscription rate by job;
- subscription rate by education;
- effect of housing loan;
- effect of personal loan;
- previous campaign outcome;
- number of previous contacts;
- account balance.

Do not create graphs merely for the sake of creating graphs.

For every important observation, explain what it could mean to the bank.

4. Data Preparation

Prepare the data appropriately for Machine Learning.

This may include:

- separating features and target;
- handling categorical variables;
- handling unknown values appropriately;
- removing variables that should not be available at prediction time;
- train/test split;
- scaling numerical variables where required.

Remember that KNN is distance based and therefore feature scaling is particularly important.

5. Build Classification Models

Build the following three models:

Model 1

Logistic Regression

Model 2

K-Nearest Neighbours

Experiment with reasonable values of K.

Model 3

Decision Tree

Experiment with appropriate parameters such as tree depth where necessary.

6. Model Evaluation

Do not select the best model based only on accuracy.

For every model, report and interpret:

- Accuracy
- Precision

- Recall
- F1 Score
- Confusion Matrix

Explain:

- True Positive
- True Negative
- False Positive
- False Negative

in the context of this bank.

For example:

False Positive:

The model says the customer is likely to subscribe, the bank calls the customer, but the customer does not subscribe.

False Negative:

The model says the customer is unlikely to subscribe, but the customer would actually have subscribed.

Discuss which type of mistake could be more important to the marketing team.

Final Business Recommendation

Assume the Head of Marketing is not a Data Scientist.

Provide a recommendation covering:

1. Which model would you recommend?
2. Why?
3. What does the model do well?
4. Where does it make mistakes?
5. Which customer groups appear more likely to subscribe?
6. How should the marketing team use the model?
7. What limitations should the business understand before deploying it?

Do not simply state:

“Decision Tree has the highest accuracy.”

Explain why the selected model makes sense **for the business**.
